

A Civic AI Assistant for SNAP Benefits

AI Chatbot for Student Screening Eligibility



Agenda

1. Problem Definition

2. Related Works

3. Methods

4. Solutions

5. Analysis

6. Impact & Future Work

01

Problem Definition

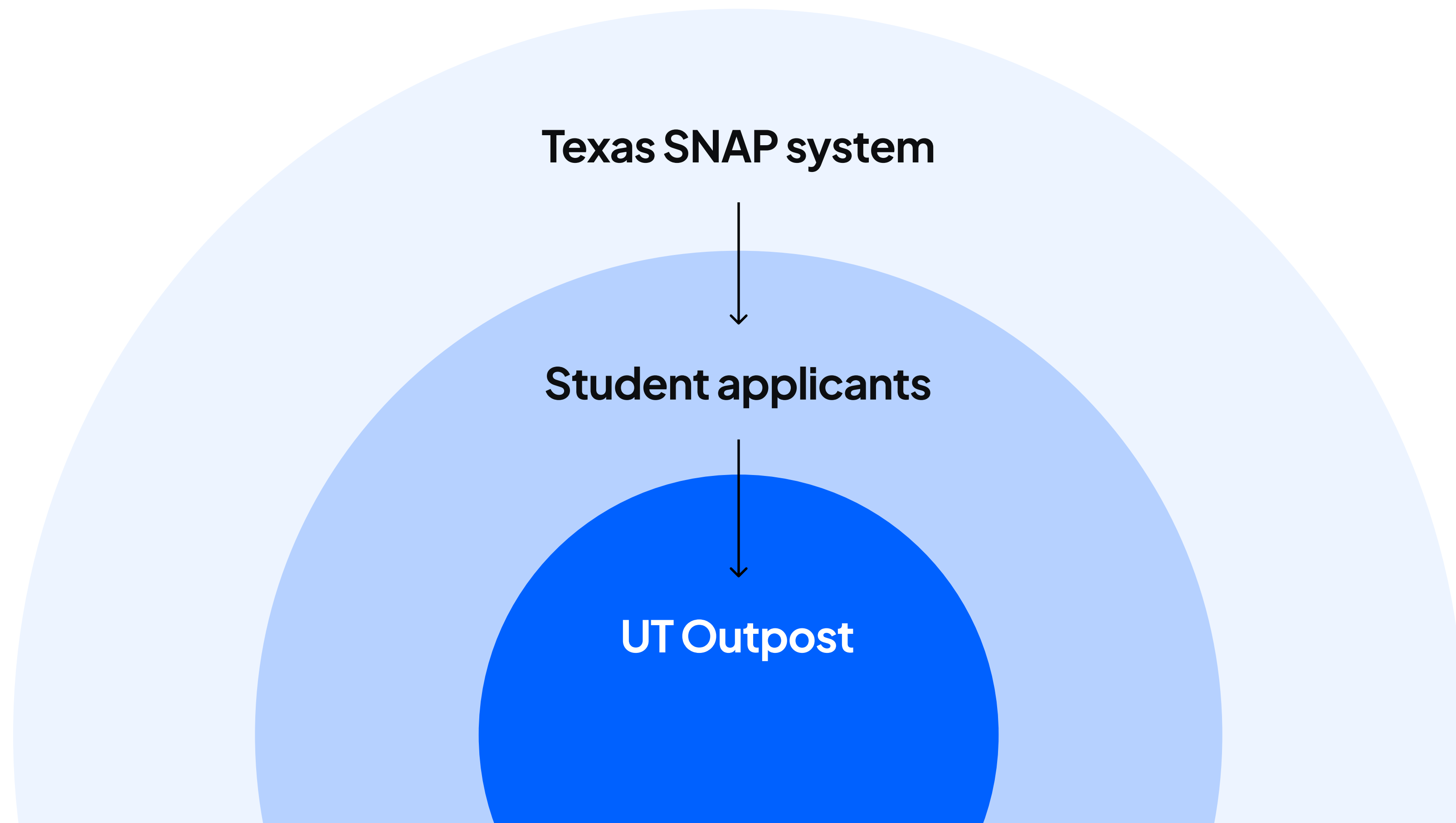


67% of eligible college students did not receive SNAP benefits.

In Texas, where SNAP participation is among the lowest nationwide, college students face even greater barriers to access.

Narrowing Our Focus to Student Applicants

Scoping down to where we can make a more actionable design opportunity



Key Findings from UT Outpost

Most college students who qualify for SNAP don't know it, and those who try to find out often give up

Finding 01

**College Students Face
Uniquely Complex
Eligibility Rules**

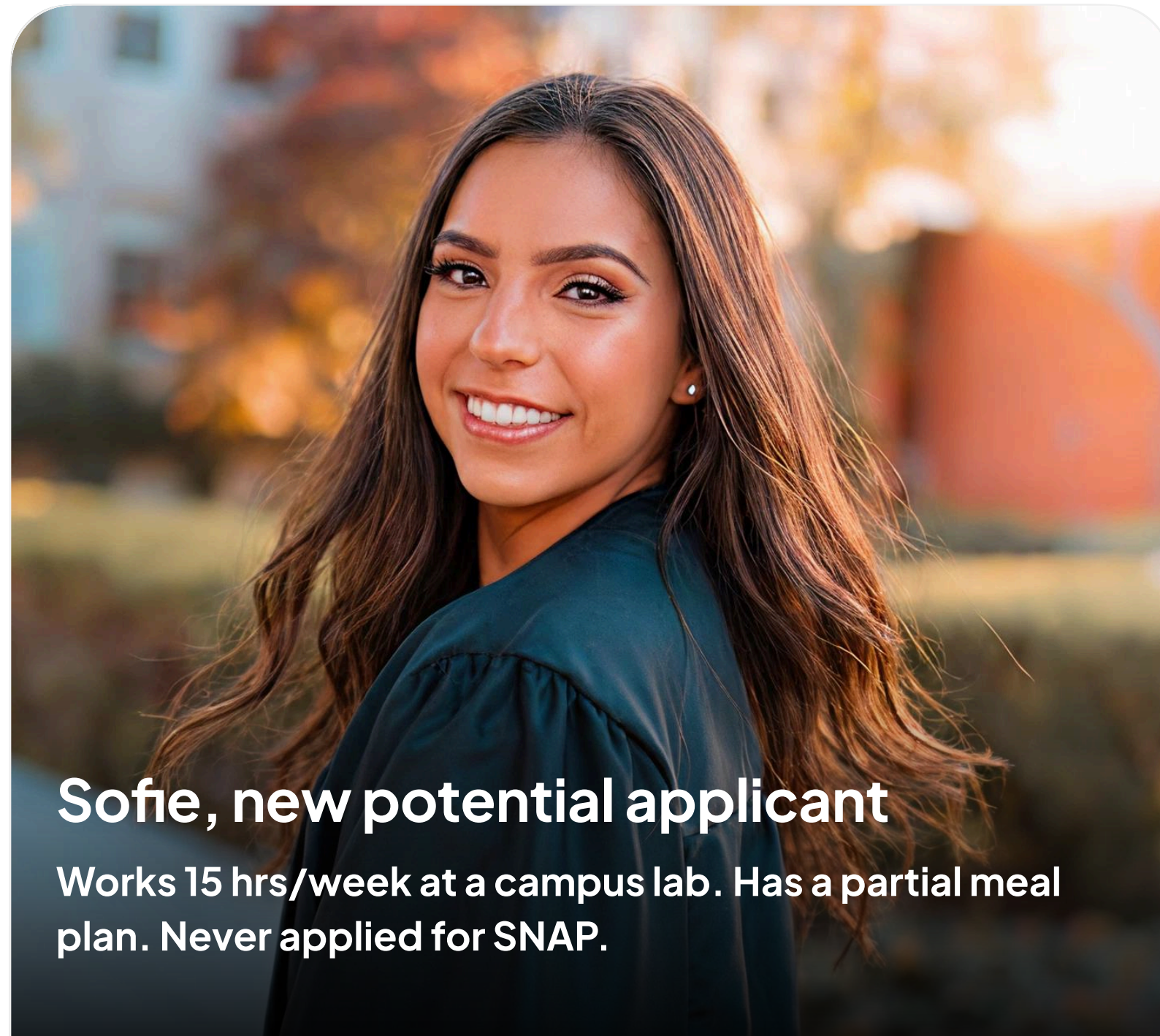
Finding 02

**Documentation &
Eligibility Are Constantly
Shifting**

Finding 03

**Privacy &
Personalization Must Be
Balanced**

Who We're Designing For?

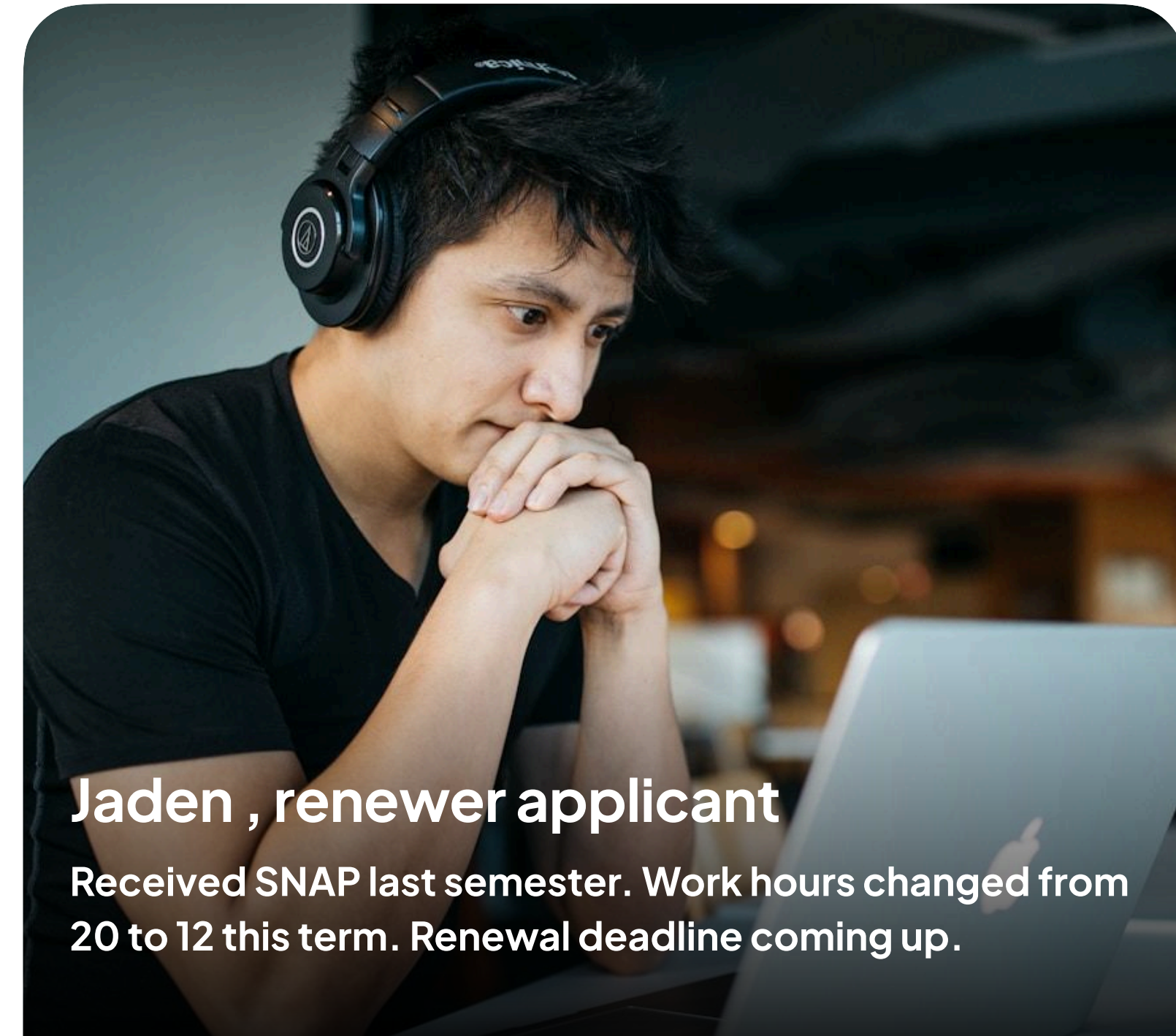


Sofie, new potential applicant

Works 15 hrs/week at a campus lab. Has a partial meal plan. Never applied for SNAP.

“

I always assumed SNAP was for families, not college students. I didn't even know students could qualify.



Jaden , renewer applicant

Received SNAP last semester. Work hours changed from 20 to 12 this term. Renewal deadline coming up.

“

My work hours dropped this semester. Last time I qualified because of my work-study. I just want to know if I still count.

02

Related Works

SBNC Eligibility Screening tool

Requires answering 33 questions to get a result

Limitation:

Linear, fixed flow

Every student must complete all questions in a fixed order, even if they only need to verify one condition.

No clarification

When students are unsure about a question or unfamiliar terms, the form offers no explanation. They can only guess.

No context or reasoning

The result shows eligible or not, but doesn't explain which rule applies or why.



**1 in 3 students qualify for up to
\$298 monthly in free groceries**

[Student Basic Needs Coalition](#) (SBNC) is a national nonprofit dedicated to tackling the food, housing, and financial insecurity crisis among college students. Since launching, we've referred students to more than \$100 million in benefits.

We encourage you to use our screener to see if you are eligible for SNAP and Medicaid.

[Check if you qualify](#)



What We Mapped from Related Works

1. Simplify the process

Only ask what's relevant to each student's situation.

Led us to design a conversational flow that adapts to each student's situation.

2. Chatbot as entry point

Chatbots have been shown to be as a low- barrier tools for civic benefit access.

Led us to choose a chatbot format over a static screener.

3. Verified sources

AI in civic contexts must be traceable, not generated from general knowledge.

Led us to prioritize transparency and accuracy in how the chatbot deliveries information.

We wanted to build a conversational AI chatbot that adapts to each student's situation, giving them a clear answer and a confident next step.

03

Methods

From 13 Steps to a 3-Phase Decision Flow

Audited the existing flow and redesigned a workflow where students only answer questions relevant to their situation

Phase 1

Quick Eligibility Check

2 - 4 questions

Student status, enrollment type, meal plan — early exit for clearly ineligible students.

Phase 2

Income Snapshot

2 - 4 questions

Household size + rough monthly income estimate against SNAP threshold.

Phase 3

Result + Take Action

Personalized output

Eligibility verdict, estimated benefit, document checklist, UT Outpost booking link.

Apply AI Capabilities

Four AI functions embedded across the three-phase eligibility flow



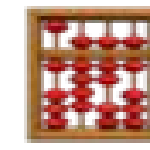
Identify

Identify user intent and route to the correct eligibility pathway.



Generate

Generate chat responses grounded in vetted policy documents



Estimate

Estimate monthly SNAP benefit amount based on household size and income



Act

Send reminder, link to application site, schedule UT Outpost advisor appointment

Ref :Yildirim, N., Oh, C., Sayar, D., Brand, K., Challa, S., Turri, V., ... & Zimmerman, J. (2023, July). Creating design resources to scaffold the ideation of AI concepts. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (pp. 2326–2346).

04

Solutions

How We Built It...

Design in Figma

Created a single key UI mockup to build visual style and component as a design reference for development.



Follow the decision flow

Based a 3-phase eligibility decision flow and designed the chatbot interaction logic.



Build with Claude Code

Fed the Figma design and decision flow to Claude Code to generate the workable prototype. The chatbot runs on Claude via the Anthropic API.



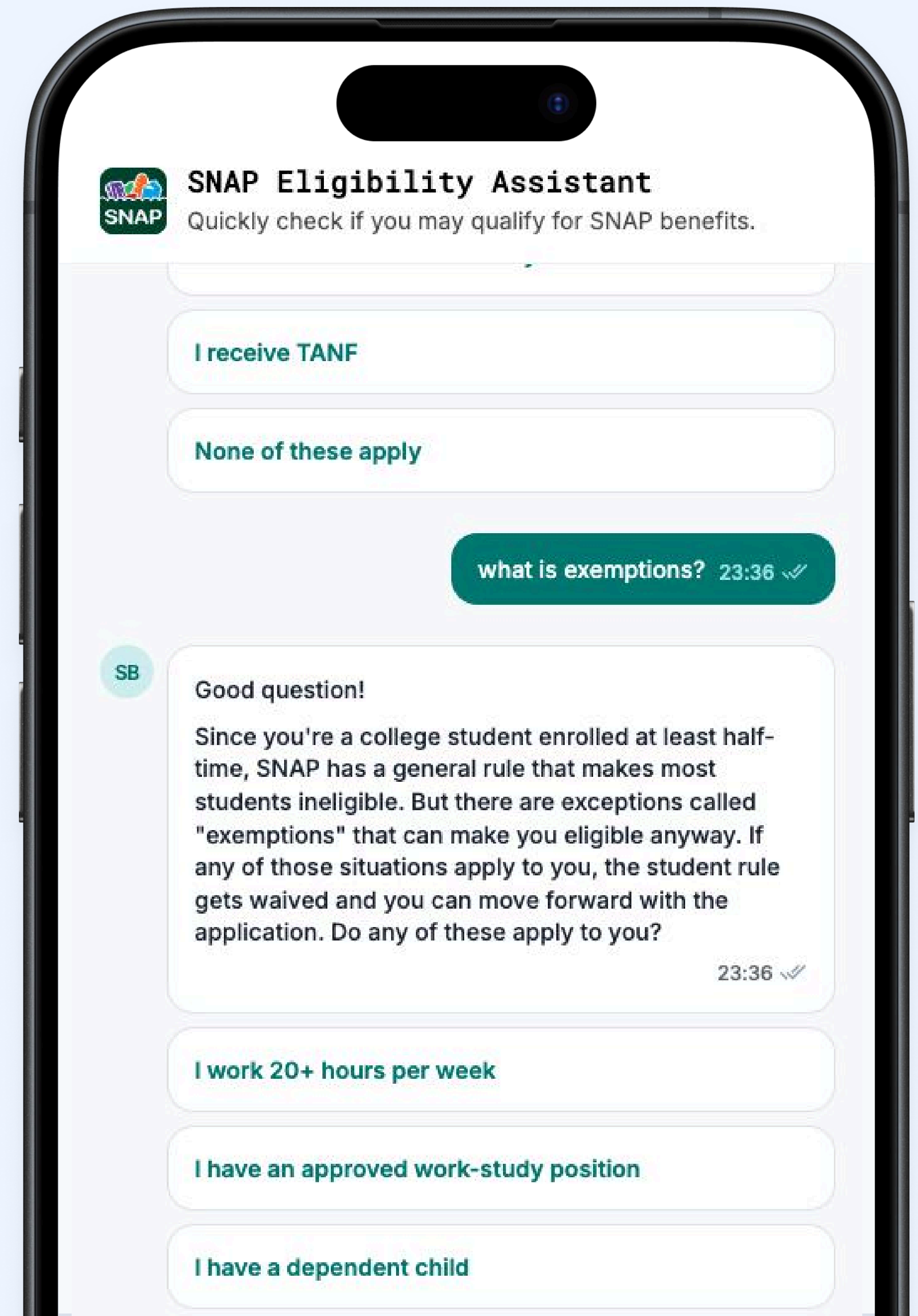
Design with AI Principles

01 Design for Mental Models

Plain language outputs

- Helps students understand if they qualify without having to read and interpret complicated policy language themselves.

Weisz, Justin D., et al. "Design principles for generative AI applications." Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems. 2024.



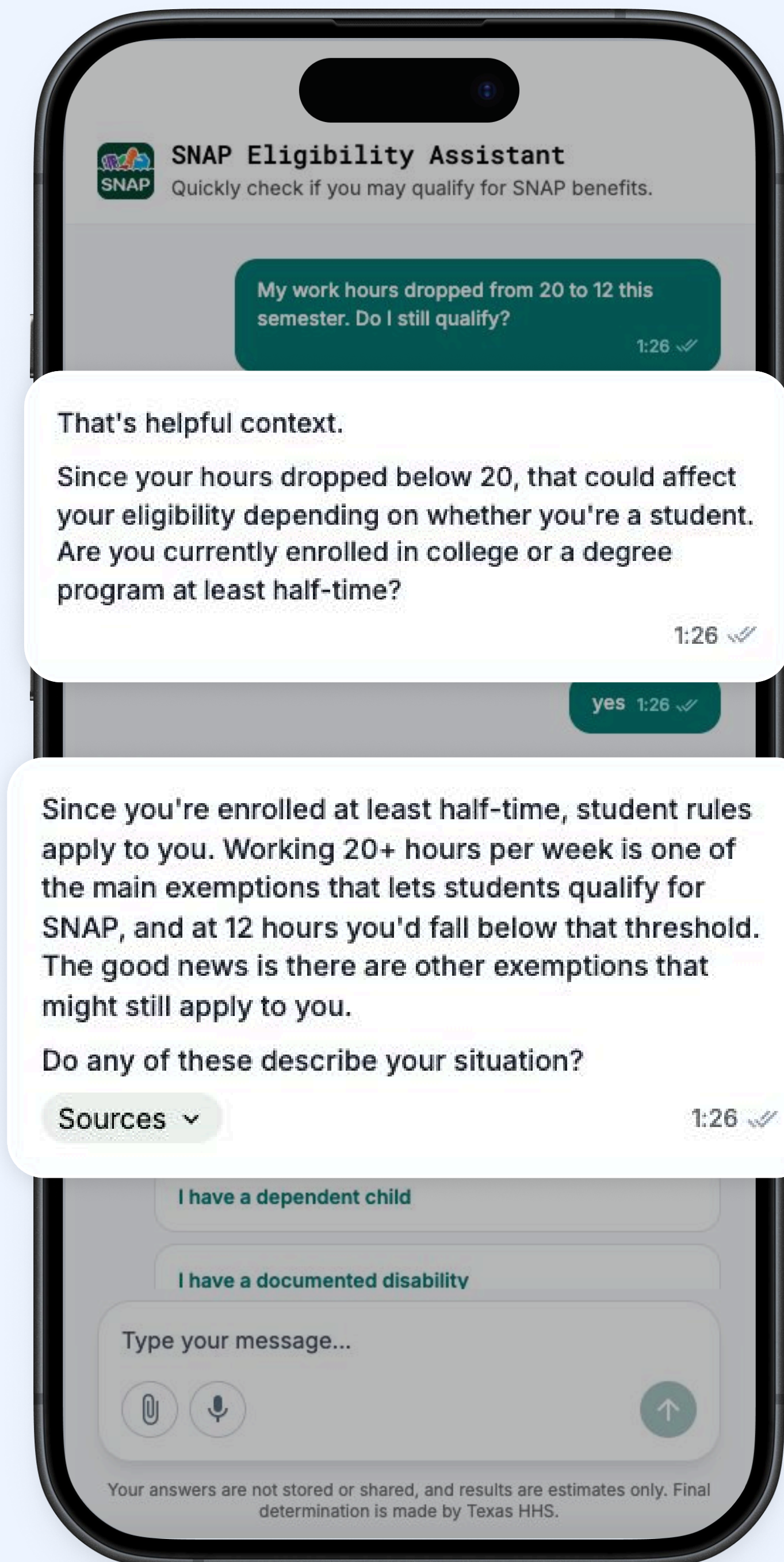
Design with AI Principles

02 Design for Co-Creation

Guided clarification questions

- Students can ask their situation directly without walking through a fixed flow and get a targeted answer.
- AI asks targeted follow-up questions to better understand the student's situation and determine their eligibility more accurately.

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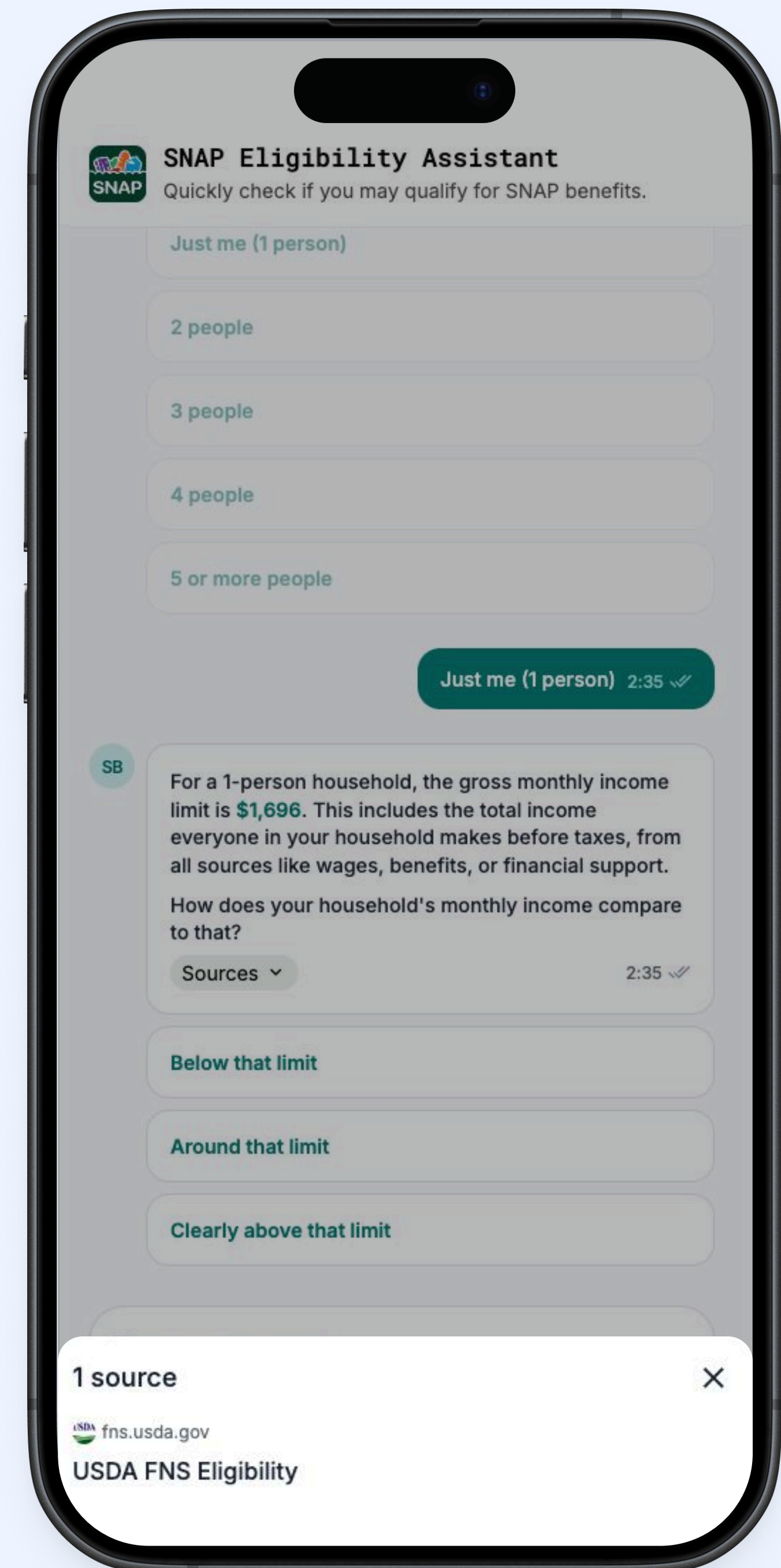
Design with AI Principles

03 Design For Appropriate Trust & Reliance

Vetted source material responses

- Policy rules, figures, and terms referenced in responses are linked to official sources.
- Students can verify any specific claim back to the policy document.

Weisz, Justin D., et al. "Design principles for generative AI applications." Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems. 2024.



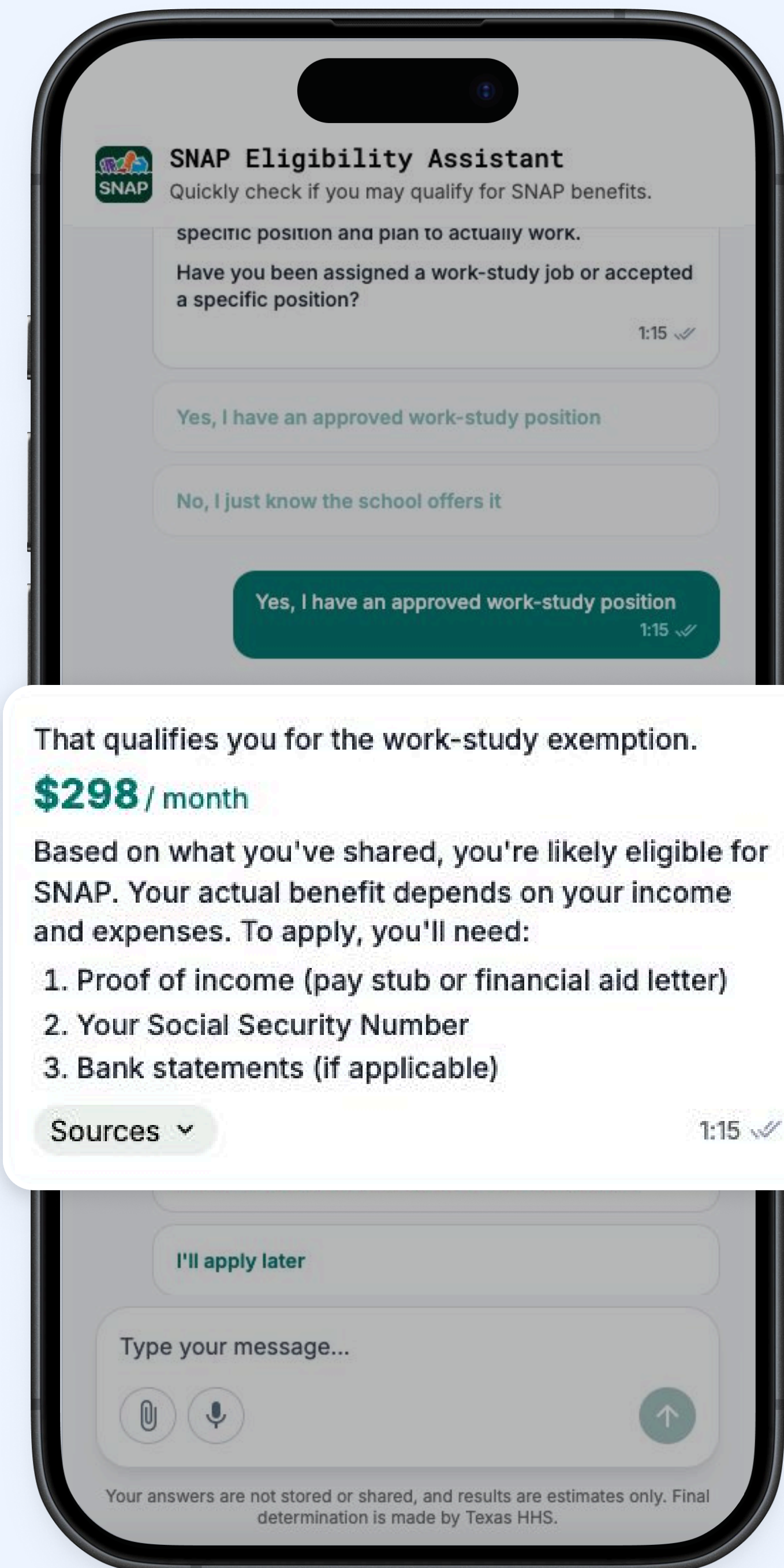
Design with AI Principles

04 Design for Imperfection

Transparent uncertainty

- Presents the results as an initial assessment, not a final decision, and concludes with a clear, specific recommendation for applicants.

Weisz, Justin D., et al. "Design principles for generative AI applications." Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems. 2024.



05

Analysis

Feedback from the users



Student Applicants

User Outcomes Metrics

- Task Completion Time
- SUS Score (system usability scale)
- Perceived Confidence & Trust

3 mins to complete eligibility check

88 SUS score

84% Perceived Trust

Tools: Post-Test Survey (Likert scale & Open-ended questions), Interviews

Ref: Microsoft HAX Toolkit HAX Design Library <https://www.microsoft.com/en-us/haxtoolkit/library/>

Feedback from the experts



UT Outpost SNAP Advisors

AI Quality Metrics

- Eligibility Accuracy
- Hallucination Rate
- Privacy Sensitivity Handling
- Social Bias / Fairness

“I like that this allows me to ask questions,
It's more **personalized** for sure”

“The eligibility results look great”

“I like the little disclaimer on the
bottom for **not storing any data** ”

“I like this **providing resources** and guides
users toward consulting with us ”

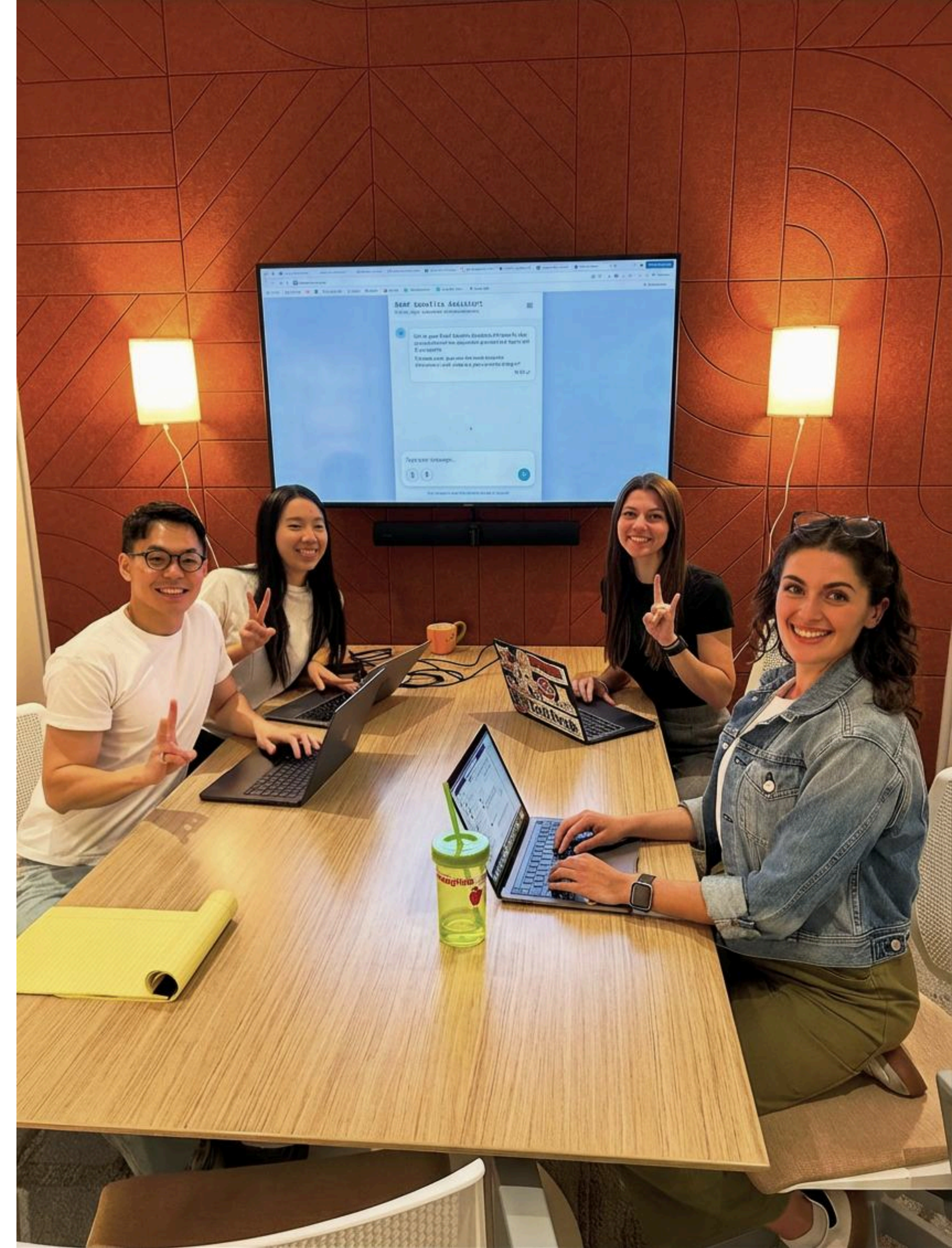
Tools: Post–Test Survey (Likert scale & Open–ended questions), Interviews

Ref: Microsoft HAX Toolkit HAX Design Library <https://www.microsoft.com/en-us/haxtoolkit/library/>

More AI, Less Humans?

Will this chatbot reduce advisor workload or replace human work?

- May **increase demand** — more eligible students book appointments
- May **reduce basic inquiries** — simple questions handled upfront



06

Impact & Future Work

Impact



SNAP Applicants

- Know if they qualify in minutes
- Ineligible students get connected to alternative food resources



**UT Outpost SNAP
Advisors**

- Easy to promote — shareable as a QR code
- Increases SNAP awareness at scale
- Frees up advisor capacity for students ready to apply

Future Work

1

Usability Testing

Collect more testing samples to better evaluate our metrics

2

Pilot & Deploy

Pilot with a broader student group and explore deployment on the UT Outpost SNAP website

3

Expand Reach

Expand beyond college students to serve the broader SNAP-eligible population

Demo / Q&A